D3.3 Homeostasis

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d3-3.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 768–778 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

The kidney

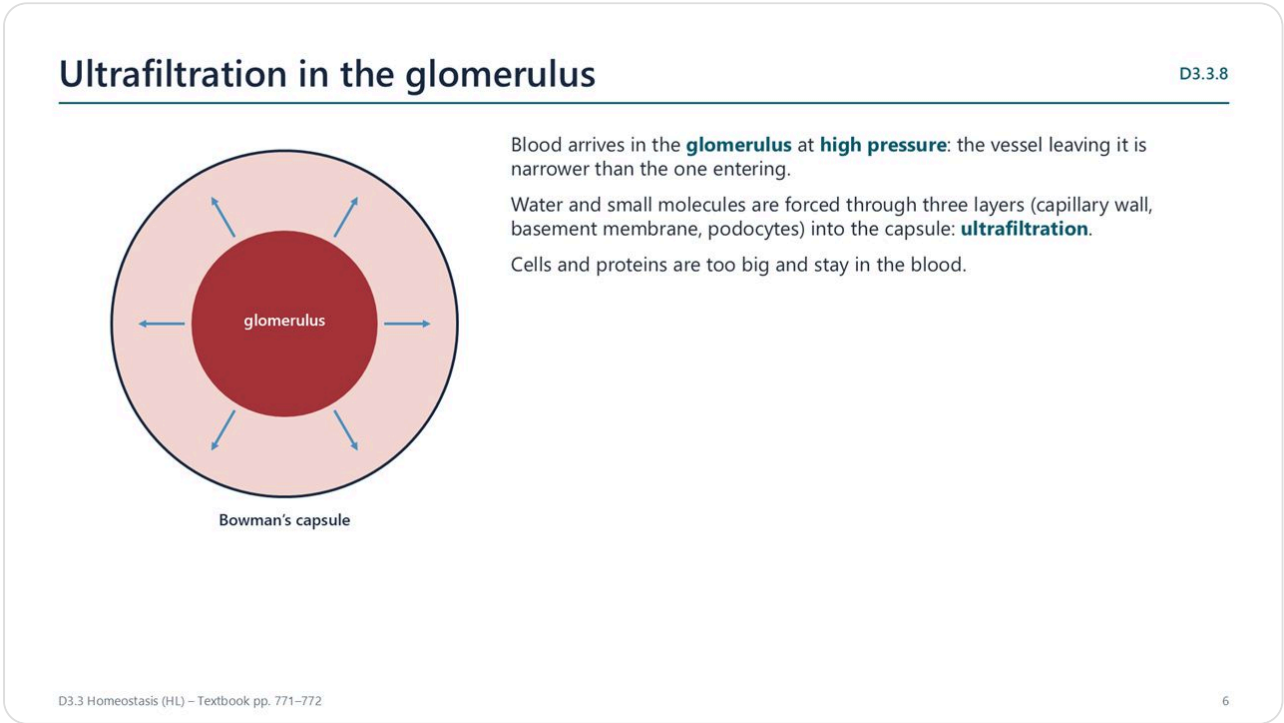
D3.3.7 Excretion and osmoregulation

pp. 768–770

Excretion removes waste products of metabolism, such as urea. **Osmoregulation** controls the osmotic concentration of body fluids, measured in osmol L^{-1}. The kidney does both: it leaves wastes in the urine and varies how much water it reabsorbs.

D3.3.8 Ultrafiltration and reabsorption

pp. 771–772



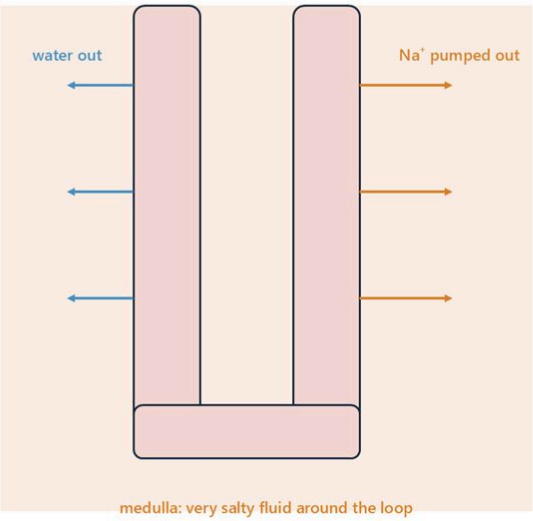
High pressure in the **glomerulus** forces plasma through fenestrated capillaries, the basement membrane and slits between **podocytes** into **Bowman's capsule**. Cells and proteins stay behind. In the **proximal convoluted tubule**, cells with microvilli and many mitochondria reabsorb all the glucose, amino acids and most salt and water, leaving urea and toxins in the filtrate.

D3.3.9 Loop of Henle

pp. 773–774

The loop of Henle

D3.3.9



The **ascending limb** pumps Na^+ out into the medulla but does not let water follow.

This makes the medulla very **salty**.

The **descending limb** lets water out by osmosis into that salty fluid.

Result: a steep salt gradient in the medulla that the collecting duct uses later.

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The ascending limb actively pumps Na^+ into the medulla but does not let water out, so the medulla becomes very concentrated. This lets water be drawn out of the collecting ducts later.

D3.3.10 ADH and the collecting duct

pp. 775–776

Osmoreceptors in the hypothalamus detect concentrated blood, and the pituitary secretes more **ADH**. ADH causes vesicles carrying **aquaporins** to fuse with the membrane of collecting duct cells, so more water is reabsorbed. When blood is dilute, ADH falls and aquaporins are taken back into vesicles.

Blood supply

D3.3.11 Blood supply and activity

pp. 777–778

Arterioles dilate or constrict to send blood where it is needed. In vigorous exercise flow to skeletal muscle rises about twentyfold while flow to the gut and kidneys falls; brain flow stays nearly constant. In sleep, flow to muscle falls.

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